

Fundamentals of Software Engineering (CS2004)

Date: February 24th 2026

Course Instructor(s)

Engr. Ansa Liaquat

Sessional-I Exam

Total Time: 1 Hours

Total Marks: 40

Total Questions: 03

Semester: SP-2026

Campus: Lahore

Dept: Data Science

Student Name

Roll No

Section

Student Signature

Vetted by

Vetter Signature

Instructions

1. You are allowed to use a one-sided, hand-written, A-4 size help sheet. Photocopies are not allowed.
2. State your assumptions clearly
3. CLO stands for Course Learning Outcome
4. CLO statements are copied from the course outline and do not help in reaching a solution.

CLO #1: Learn the basics of software process models and choose the best fit for variable nature of projects in industry

Q1: List the process model that will be the most appropriate for the following situation. Also, list (as numbered bullets) your reason(s) for choosing a particular model. If you just list the process model without mentioning the reason(s), you will not be awarded any marks. Do not provide more than 4 reasons. No reason shall take more than one line of the answer sheet. Do not write paragraphs.

Part (a) A university consortium is developing an online academic administration system to be used across multiple campuses. The client wants to see working features early and expects requirements to evolve as feedback comes from different departments. Development is divided into two-week cycles, each producing a usable increment of the system. A client representative maintains and prioritizes a list of requirements. Team members meet briefly every day to synchronize work, demonstrate completed features at the end of each cycle, and reflect on process improvements. A

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facilitator supports the team by removing obstacles, while functional software is treated as the primary measure of progress.

Part (b) A small startup is creating a **mobile health-tracking application** in a highly competitive market. Customer needs change frequently based on user feedback and analytics. Developers work in pairs, write automated tests before implementing features, integrate code several times a day, and release updates frequently. The customer works closely with the team, clarifying requirements continuously. Minimal documentation is maintained, as the team prioritizes clean code, fast delivery, and adaptability to change.

Part (c) A financial institution is developing an **intelligent fraud-detection system** for online transactions. The project involves high uncertainty due to new technologies, strict security requirements, and potentially severe financial losses if errors occur. Development proceeds in repeated cycles, each beginning with identifying and analyzing risks. Prototypes are built to reduce technical uncertainty, and stakeholders review progress and risks at the end of every cycle before deciding whether and how to proceed further.

Part (d) A public-sector organization is developing a **nationwide taxation management system**. Legal rules, tax slabs, reporting formats, and audit requirements are already defined in official policy documents. Before development begins, analysts spend several months gathering and documenting requirements, which must be formally approved by senior management. Once approved, changes are discouraged and require extensive authorization. System design is completed in full before coding starts, and developers follow this design strictly. Testing is performed after implementation is finished, and the system is deployed in one complete release. Emphasis is placed on documentation, predictability, and regulatory compliance rather than flexibility.

[5+5+5+5=20 Marks]

CLO # 5: Learn basic project management skills along with costing and estimation

Q2: The table below defines the activities within a small project.

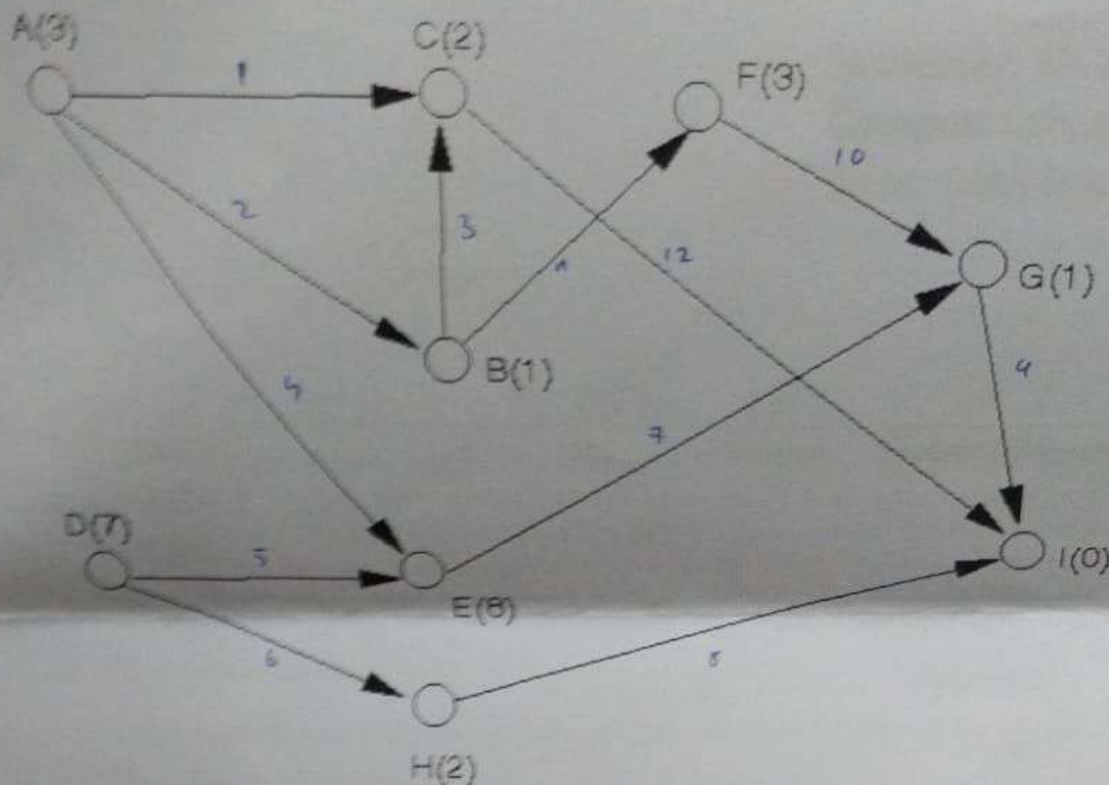
Activity	Completion time (weeks)	Immediate predecessor activities
A	2	-
B	3	-
C	4	A
D	3	B,C
E	8	A,D
F	3	C
G	2	E
H	3	F
I	6	G
J	7	H
K	5	I,J

To do: Draw the activity on node graph.

[10 marks]

CLO # 5: Learn basic project management skills along with costing and estimation

Q3: Consider the following activity on node graph where duration of activity is mentioned in days (brackets). [10 marks]



To Do: Calculate the minimum overall project completion time and identify which path and activities are critical. What is the slack time associated with each of the non-critical activities? Calculate earliest and latest start time and finish time as well.